

Operational Intelligence Evidence Pack

Benchmark rubric, review model, control comparisons, and falsification criteria for Operational Intelligence.

Evidence Question

What evidence would convince another experienced engineer that this operating model is useful? Operational Intelligence becomes stronger when claims can be tested, challenged, compared, and revised.

Review path

Structured practitioner review is captured through /contact so SRE, architecture, AI engineering, governance, executive, and product reviewers can challenge the doctrine with specific evidence requests.

Evidence Classes

Class	Purpose	Strong signal
Implementation example	Show end-to-end behavior	Public-safe inputs, outputs, states, and decisions
Benchmark fixture	Make behavior repeatable	Prompt, context, expected behavior, refusal, citation
Control comparison	Separate value from baselines	Dashboard-only, chatbot-only, ticket-only, and OI workflow compared
Practitioner review	Test real-world comprehension	Structured review by SRE, architect, AI engineer, governance, executive
Regression history	Show maturity over time	Failures tracked and corrected
Negative case	Test boundaries	Refuses confidential, unsupported, or unsafe requests

OI-ROOM-001 Benchmark Rubric

Dimension	Acceptance question
Retrieval	Did the system retrieve doctrine, reference architecture, and relevant OI-ROOM-001 context?
Grounding	Are all material claims supported by approved public content?
Evidence attribution	Are observation, inference, contradiction, missing evidence, and confirmed fact separated?
Hypothesis lifecycle	Do hypotheses move through explicit states?
Contradiction handling	Does contradictory evidence change confidence or route?
Decision safety	Are actions framed as reviewable options?
Operator control	Are consequential actions gated by explicit human approval?
Replay reproducibility	Can the scenario be replayed from approved context?
Refusal	Does the system refuse confidential or unsupported questions?

Evidence Ledger

Claim	Classification	Result	Next improvement
OI focuses on the decision path from evidence to action.	Original synthesis	Mixed	Collect SRE and observability practitioner feedback
The ten-layer model can guide implementation.	Original synthesis	Not independently tested	Review a second public-safe implementation example
Evidence graphs improve reviewability.	Derived	Plausible	Compare against chatbot-only and dashboard-only explanations
Replay seeds support regression testing.	Derived	Partial	Add replay-backed workflow fixtures
Operator control should gate consequential actions.	Established governance principle applied to OI	Strong	Add approval class examples

Minimum Conformance Checklist

Requirement	Observable proof	Failure signal
Evidence before conclusion	Recommendations cite approved evidence or missing evidence.	Fluent RCA with no source trail.
Transaction context	Affected customer, process, or business journey is identified.	Local metric treated as the whole incident.
Topology boundary	Dependency, ownership, blast radius, and freshness are visible.	Stale service map treated as fact.
Evidence typing	Observation, inference, contradiction, missing evidence, and fact are separated.	Contradictions hidden in prose.
Hypothesis lifecycle	Competing hypotheses move through explicit states.	First plausible explanation becomes final answer.
Evaluation gate	Grounding, citations, refusal, contradictions, and action safety are tested separately.	Single aggregate score implies trust.
Decision packet	Action includes risk, reversibility, owner, fallback, and approval class.	Assistant recommends action without review context.
Operator control	Approval, escalation, override, and refusal boundaries are explicit.	Automation boundary is hidden.
Replay seed	Scenario can be reproduced from approved context and expected behavior.	Demo cannot be rerun after changes.
Learning loop	Reviewed outcome updates memory, patterns, docs, and fixtures.	Learning remains unstructured narrative.

Weakening Conditions

The doctrine should be revised if experienced SRE teams cannot distinguish it from existing practice, if independent teams interpret the layers incompatibly, if evidence graphs add complexity without clarity, or if evaluation gates fail to catch obvious regressions.